Technical Documentation

RAG-Based Customer Support Assistant using LangGraph and Human-in-the-Loop

---

1. Introduction

What is RAG

Retrieval-Augmented Generation (RAG) is an architecture that combines information retrieval with large language models (LLMs). Instead of generating answers purely from pre-trained knowledge, the system retrieves relevant information from an external knowledge base and injects it into the prompt before generating a response.

Why it is Needed

Traditional LLMs suffer from hallucination and lack access to domain-specific or updated information. RAG solves this by grounding responses in real data, improving accuracy and reliability.

Use Case Overview

This system is designed as a customer support assistant that:

- Reads company PDF documents
- Retrieves relevant information
- Answers user queries accurately
- Escalates complex or uncertain queries to human agents

---

2. System Architecture Explanation

The system follows a modular RAG architecture consisting of two main pipelines:

Ingestion Pipeline

PDF → Text Extraction → Chunking → Embedding → ChromaDB

Query Processing Pipeline

User Query → LangGraph Workflow → Intent Detection → Retrieval → LLM → Response → HITL (if needed)

Component Interactions

- The user interacts via UI
- Query enters LangGraph workflow
- Intent detection determines routing
- Retriever fetches relevant chunks from ChromaDB
- LLM generates response using retrieved context
- Decision layer evaluates confidence
- HITL handles escalation if required

---

3. Design Decisions

Chunk Size Choice

- 500–1000 tokens with overlap of 100–200
- Ensures balance between context retention and retrieval precision

Embedding Strategy

- Dense vector embeddings using OpenAI or Sentence Transformers
- Captures semantic similarity between queries and documents

Retrieval Approach

- Top-K similarity search using cosine similarity
- Returns most relevant chunks for context

Prompt Design Logic

- Combine user query + retrieved chunks
- Use structured prompts to guide LLM
- Include instructions to avoid hallucination

---

4. Workflow Explanation

LangGraph Usage

LangGraph is used to define the system as a directed graph where nodes represent processing steps and edges define transitions.

Node Responsibilities

- Input Node: Accept user query
- Processing Node: Detect intent and retrieve data
- LLM Node: Generate response

- Decision Node: Evaluate confidence
- HITL Node: Handle escalation
- Output Node: Return response

State Transitions

The state object flows through nodes and is updated at each stage:

- Query added at input
- Intent added at processing
- Retrieved chunks added before LLM
- Response and confidence added after LLM

---

5. Conditional Logic

Intent Detection

- Classifies queries into:
  - FAQ
  - Complex
  - Escalation

Routing Decisions

- FAQ → Direct retrieval + answer
- Complex → Retrieval + refinement
- Escalation → HITL

---

6. HITL Implementation

Role of Human Intervention

- Handles low-confidence or sensitive queries
- Ensures correctness where AI is uncertain

Benefits

- Improves reliability
- Builds user trust
- Reduces incorrect responses

Limitations

- Increased response time
- Requires human resources

7. Challenges & Trade-offs

Retrieval Accuracy vs Speed

- More chunks improve accuracy but increase latency

Chunk Size vs Context Quality

- Larger chunks retain context but reduce precision

Cost vs Performance

- Better models increase cost but improve quality

---

8. Testing Strategy

Testing Approach

- Unit testing for modules
- Integration testing for pipeline
- Manual testing with sample queries

Sample Queries

- “What is the refund policy?”
- “How do I claim warranty?”
- “I have a complaint about service”

---

9. Future Enhancements

- Multi-document support
- Feedback loop for learning
- Conversation memory integration
- Deployment using cloud infrastructure

---

10. Conclusion

This system demonstrates how RAG can be used to build reliable AI-powered support assistants. By combining retrieval, generation, workflow orchestration, and human intervention, the system achieves accuracy, scalability, and practical usability.